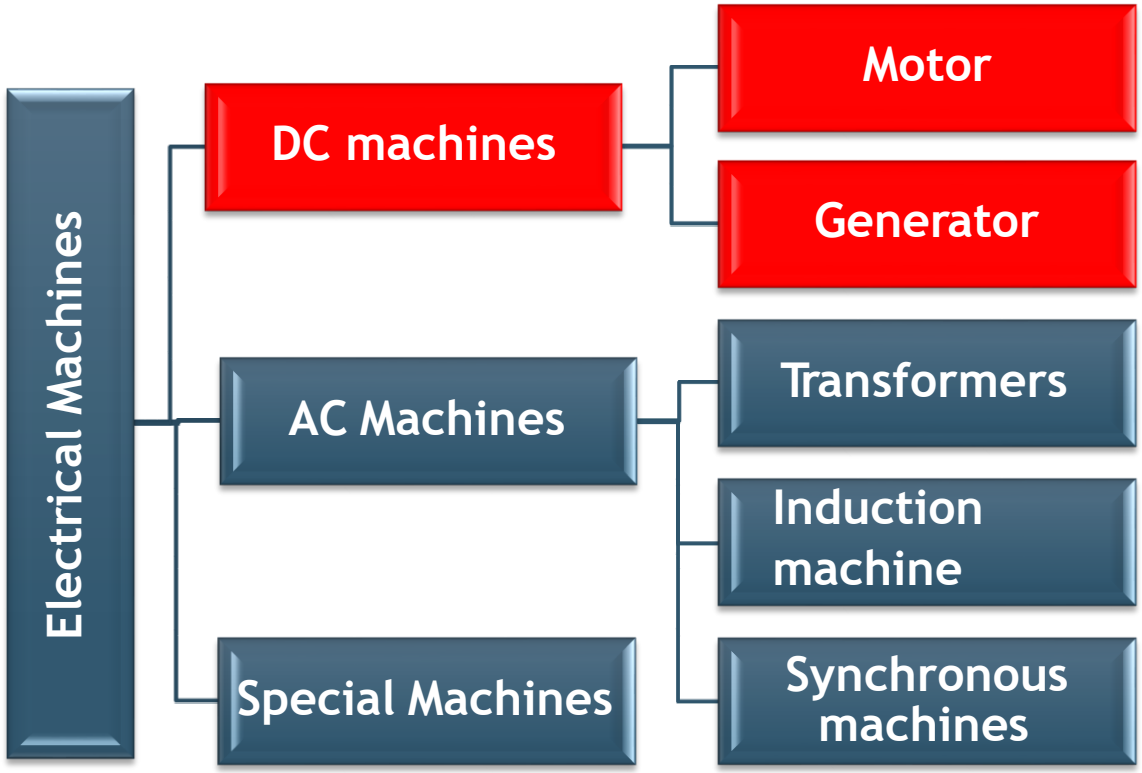
ELECTRICAL MACHINES

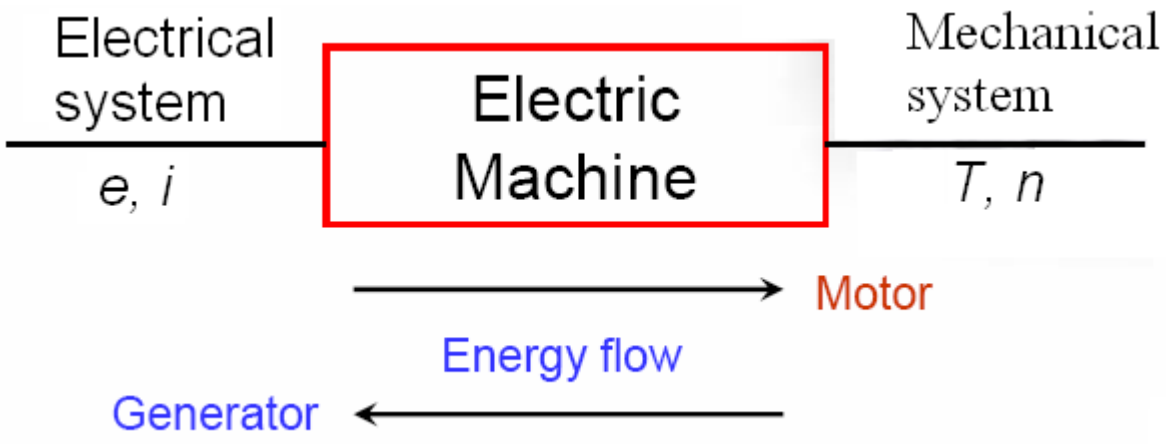
ELNG 212

By: Joshua Dantuo

Introduction

Faraday's Law





Machines are called

- AC machines (generators or motors) if the electrical system is **AC**.
- DC machines (generators or motors) if the electrical system is **DC**.

Direct Current (DC) Machines Fundamentals

Lets formulate this in a more "scientific way"

- **Generator action**: An emf (voltage) is induced in a conductor if it **moves through a magnetic field.**
- **Motor action**: A force is induced in a conductor that has a current going through it and **placed in a magnetic field.**

Any DC machine can act either as a generator or as a motor. Not all machines have this feature except for the DC machine

Applications of DC Motors:

1. D.C Shunt Motors:

It is a **constant speed motor**. Where the speed is required to remain almost constant from no-load to full load. Where the load has to be driven at a number of speeds and any one of which is nearly constant.

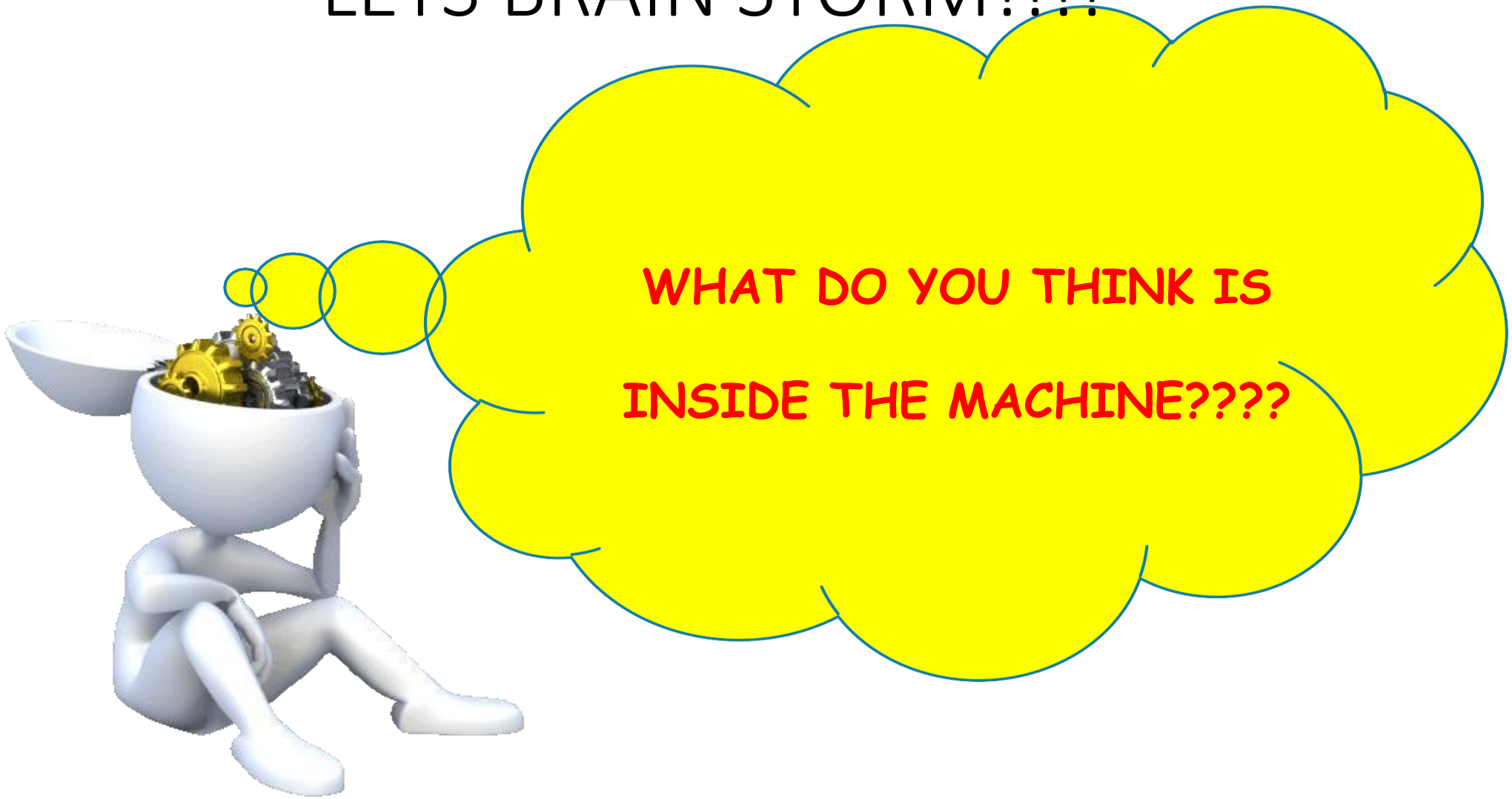
- Lathes
- Drills
- Boring mills
- Shapers
- Spinning and Weaving machines.

2. D.C Series motor:

It is a **variable speed motor. The speed is low at high torque.** At light or no load ,the motor speed attains dangerously high speed.

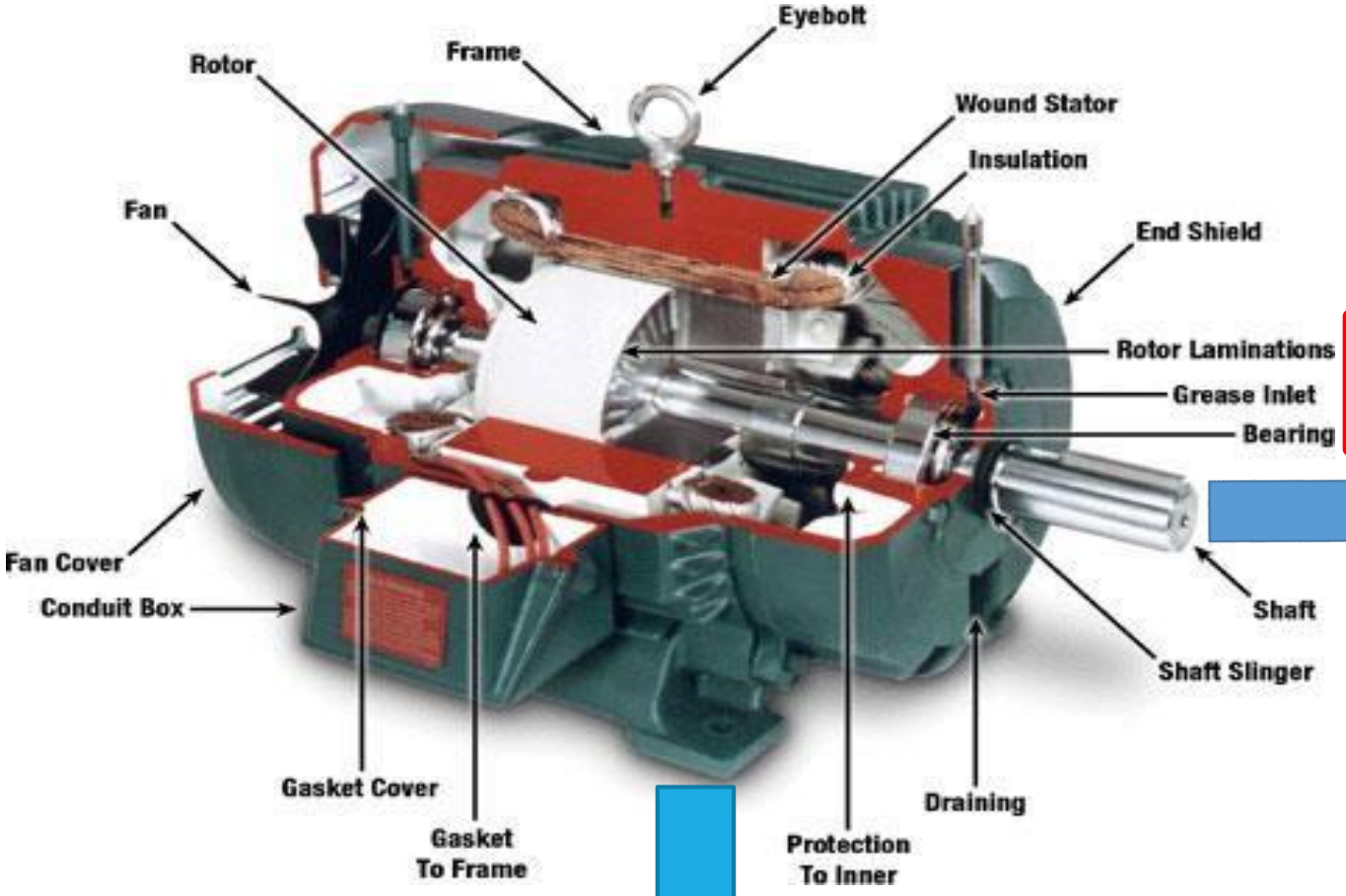
- Electric traction
- Cranes
- Elevators
- Air compressor
- Vacuum cleaner
- Hair drier
- Sewing machine

LETS BRAIN STORM!!!!

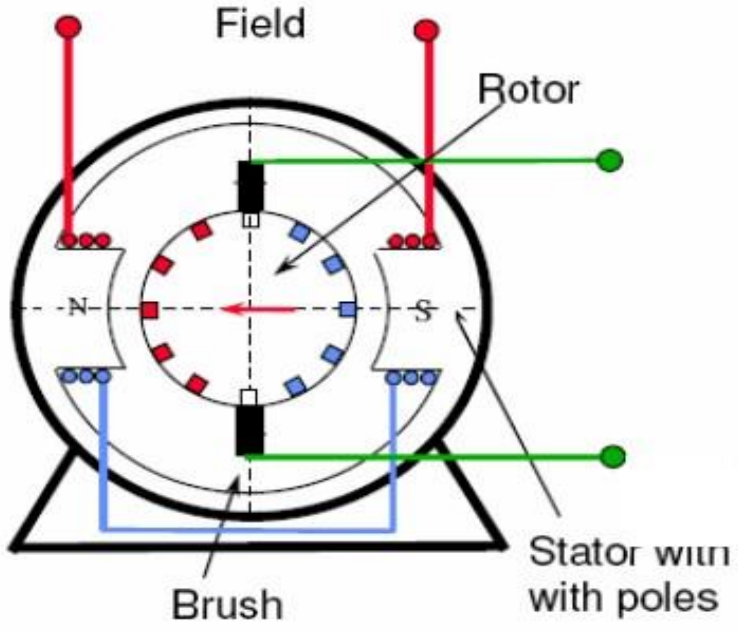


Construction of DC machine

Two electrical circuits present in the dc machine:
1- Field circuit
2- Armature circuit

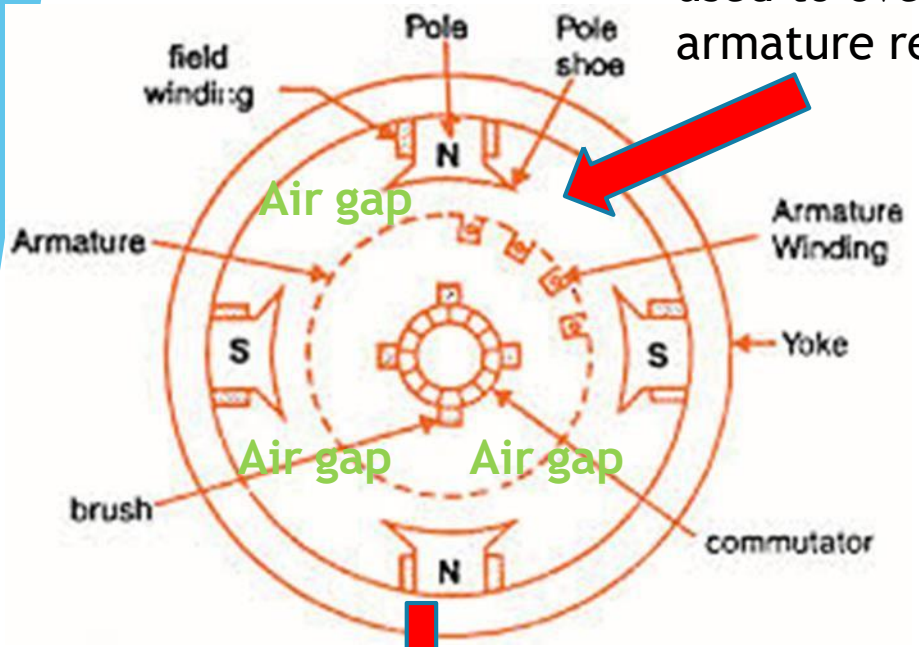


Rotor: rotating part of the machine



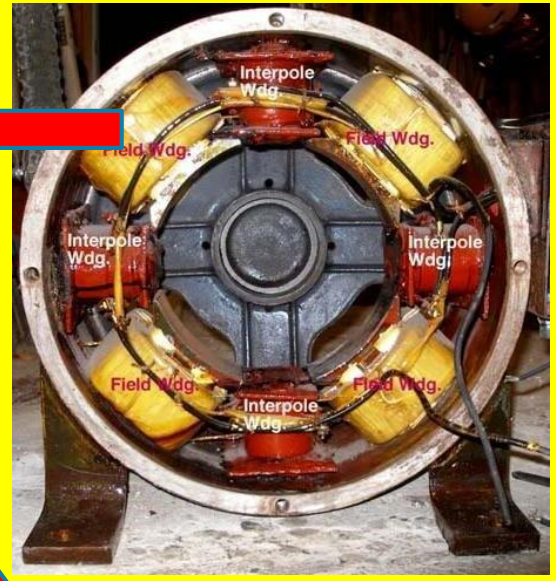
Stator: Stationary part of the machine

1- Stator:



Inter Poles: located between poles and used to overcome armature reaction

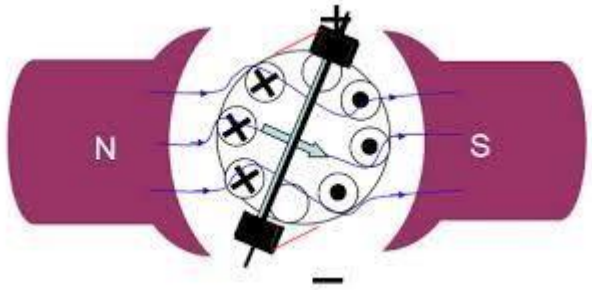
Poles: projects inwards and provides a path for the magnetic flux



THE STATOR COULD BE LAMINATED OR MADE OF SINGLE CAST PIECE OF METAL

Poles: the end of the poles that are close to rotor "spread out" over the rotor surface to distribute flux evenly over the rotor surface. We call the end as "pole shoe". Due to their spread out they are often called Salient Poles.

Frame: provides physical support



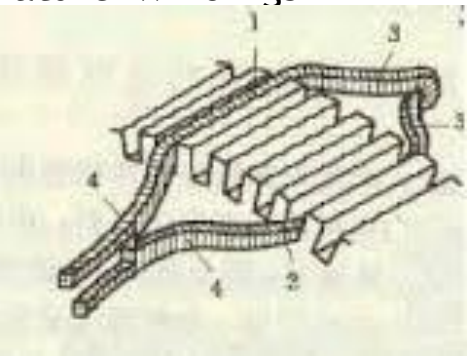
Field windings: windings responsible magnetic flux production

Stator

2- Rotor: Rotating part of machine

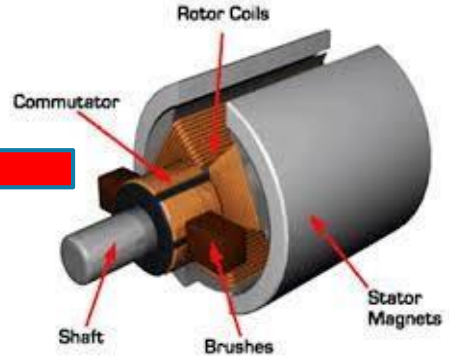
Rotor of dc machine is often called "armature" as it holds the armature windings

Armature winding: carries current crossing the field, thus creating shaft torque in a rotating machine or force in a linear machine as well as generate an electromotive force (EMF). Some call it "The power-producing component" of an alternator, generator, dynamo or motor.

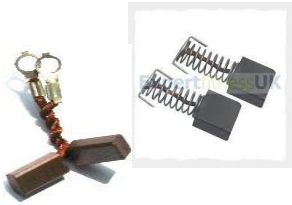


Armature

Commutator: built on the shaft of the rotor at one end of the core. Made of copper bars insulated by mica . Mica is very hard and is harder than the commutator material itself for good sticking. Serves as a "mechanical rectifier".



Brushes: made of carbon, graphite or a mixture of both. They have high CONDUCTIVITY and low friction coefficient to reduce the wear but they are softer than commutator to avoid commutator wear. It is very much affected by the current flowing in them and how they are adjusted.

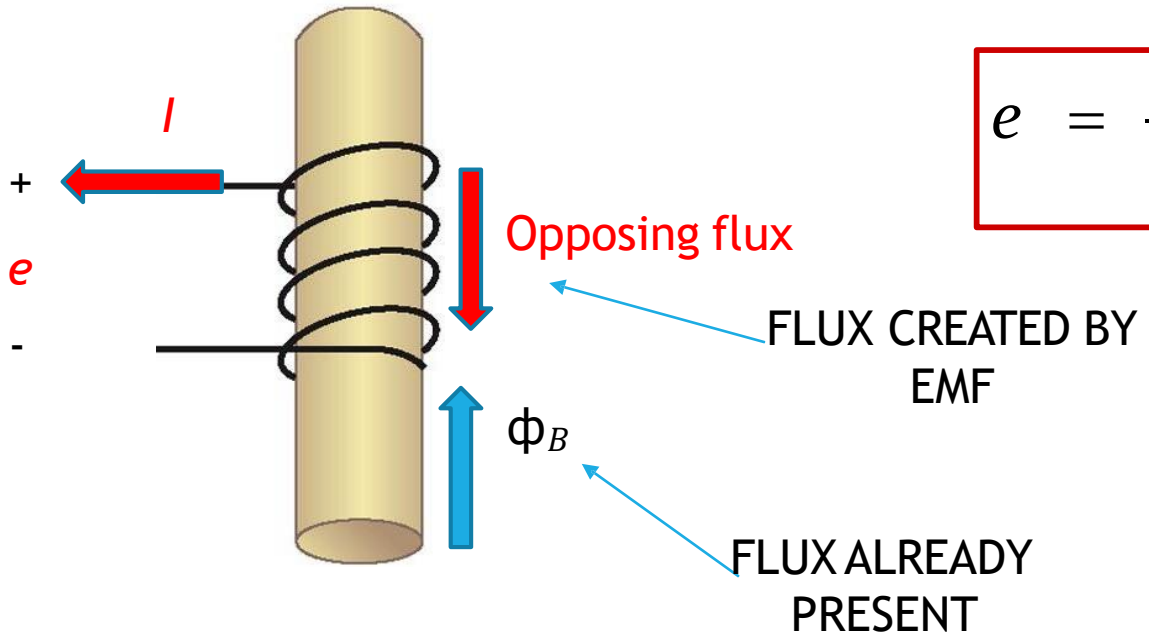


THE ROTOR IS COMPOSED OF MANY LAMINATIONS STAMPED FROM A STEEL PLATE.

Faraday Laws

HOW CAN MAGNETIC FIELD AFFECT THE SURROUNDING

1- If a flux passes through a turn of coil of a wire, a voltage will be induced in the turn of wire that is directly proportional to the *rate of change in flux with respect to time*.



$$e = -N \frac{\Delta\Phi}{\Delta t}$$

e= average emf (V)
N= number of turns
ϕ = flux passing through the turn
t= time

-ve sign is an expression of Lenz's law: The direction of the voltage buildup in a coil is such that if the coil end were short cct, it would produce current that would cause a *flux opposing* the original flux change

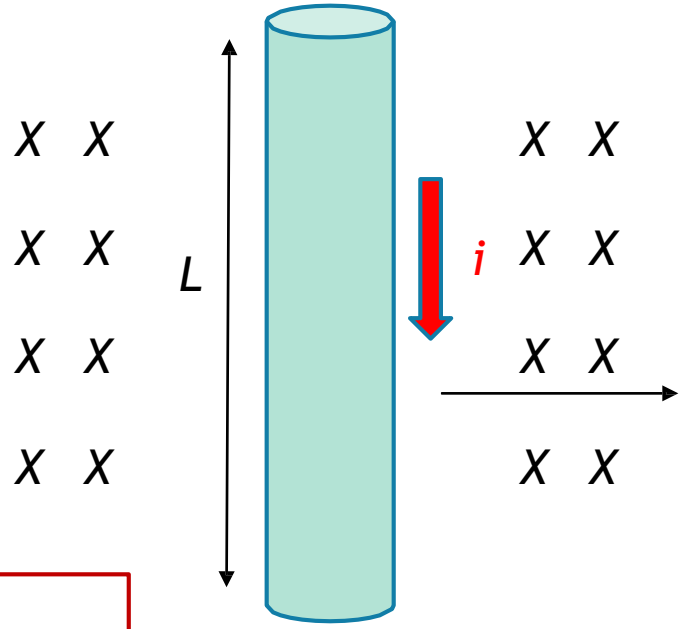
If a flux is *increasing* in strength, then the voltage built up in the coil will tend to establish a flux that will oppose the increase

Faraday Laws

HOW CAN MAGNETIC FIELD AFFECT THE SURROUNDING

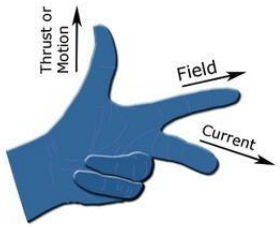
2- Magnetic field induces a force on a current carrying wire within the field.

Field into the page



B= magnetic flux density (wb/m²)
i= current (A)
F= force induced (N)
L= length of conductor (m)

$$F = i(L \times B)$$



MOTOR ACTION

$$F = iLB\sin\Theta$$

Θ = angle between the wire and the flux density vector

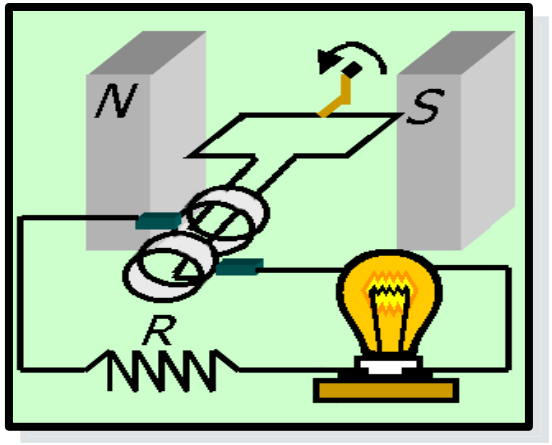
$$B = \frac{\Phi}{A}; \quad \Phi = BA$$

Force direction is given by the ***left-hand*** *rule*

The induction of a force in a wire by a current in the presence of a magnetic field is the basis of the *motor action*.

The EMF equation :

Assume one coil only now



Let,

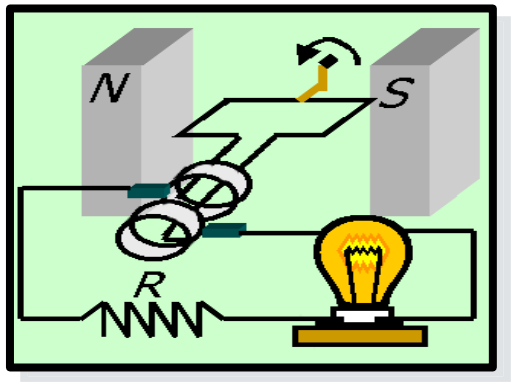
- ϕ= flux per pole in weber
- Z = Total number of conductor ← One coil = 2 conductor
- P = Number of poles
- a = Number of parallel paths: This describes the way the machine's armature conductors are connected relative to each other and to the number of poles. The two basic ways of connecting these conductors are called '**lap**' and '**wave**', but it gets more complicated.

  a=P (lap)

  a=2 (wave)
- n =armature speed in rpm
- e = emf generated in any one of the parallel path

The EMF equation :

EMF is induced in the conductor according to Faraday's law.



The average value of e.m.f. induced in each armature conductor is,

$$e = -N\frac{d\phi}{dt}$$

Consider one revolution of conductor. In one revolution, conductor will cut total flux produced by all the poles i.e. ϕ * P.

- The time required to complete one revolution is 60/n seconds as speed is n r.p.m. Hint: rpm (revolutions per minute)

n (rev) 1 min * 60 (sec)
1 rev ????? (sec)

EMF produced by one conductor

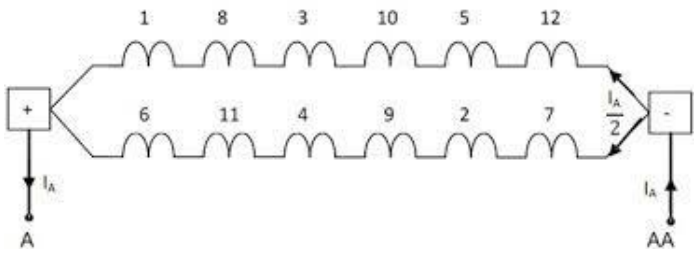
$$e_{cond} = \frac{\phi P}{\frac{60}{n}} = \phi P \frac{n}{60}$$

Numerator=$N\,d\varphi$

denomenator= dt

$e = -N\frac{d\phi}{dt}$= 1 * change of flux / time

Now the conductors in one parallel path are always in series. There are total Z conductor with *a* parallel paths, hence Z/a number of conductors are always in series and e.m.f. remains same across all the parallel paths.



Total EMF produced by armature conductors

$$e_{total} = \phi P \frac{n}{60} \times \frac{Z}{a}$$

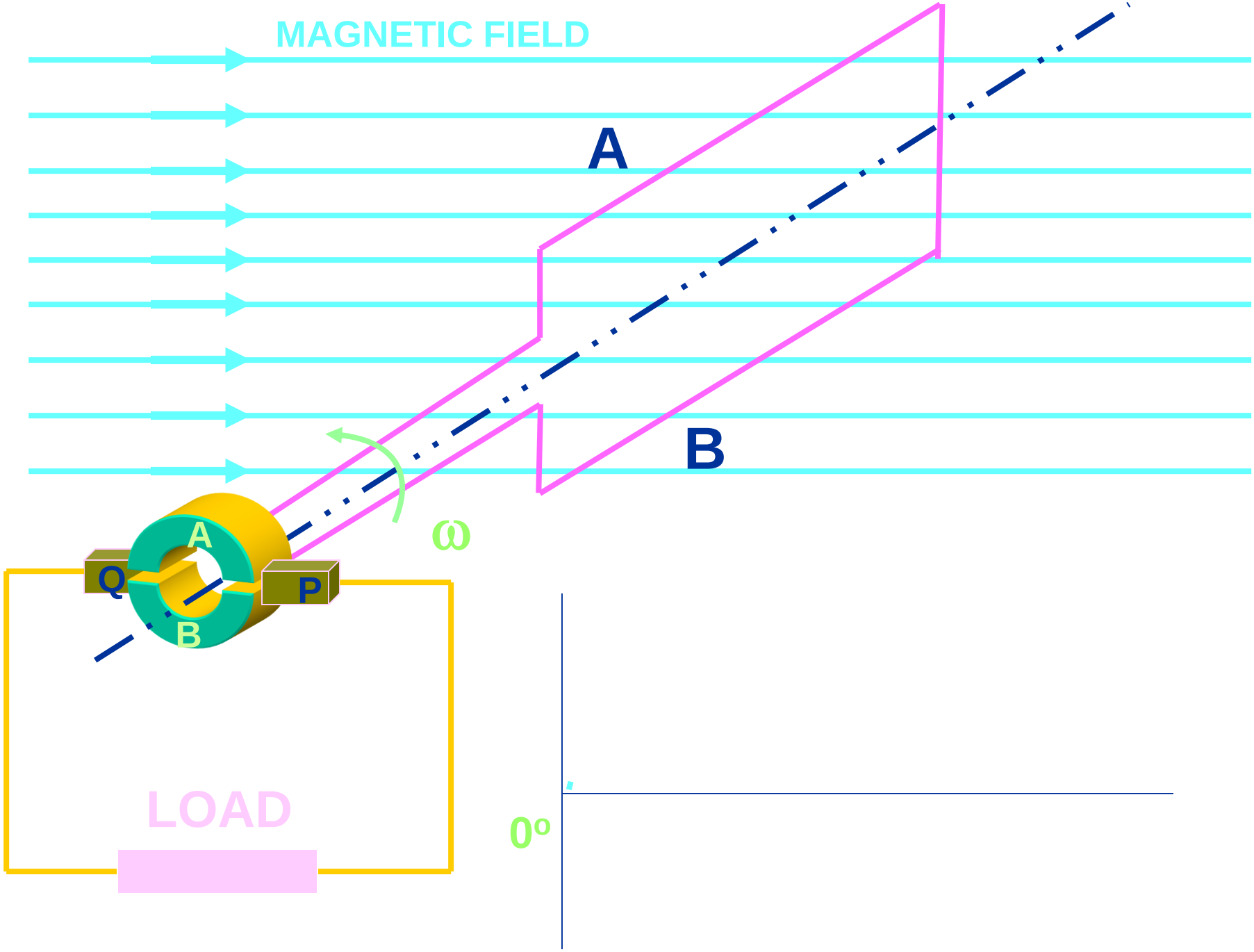
P, Z, a: design parameters

N, ϕ: control parameters

MOST IMPORTANT EQUATION IN DC GENERATORS

$$e_{total} \propto k\,\phi\,n$$

EMF is proportional to the field and speed of rotation


MAGNETIC FIELD
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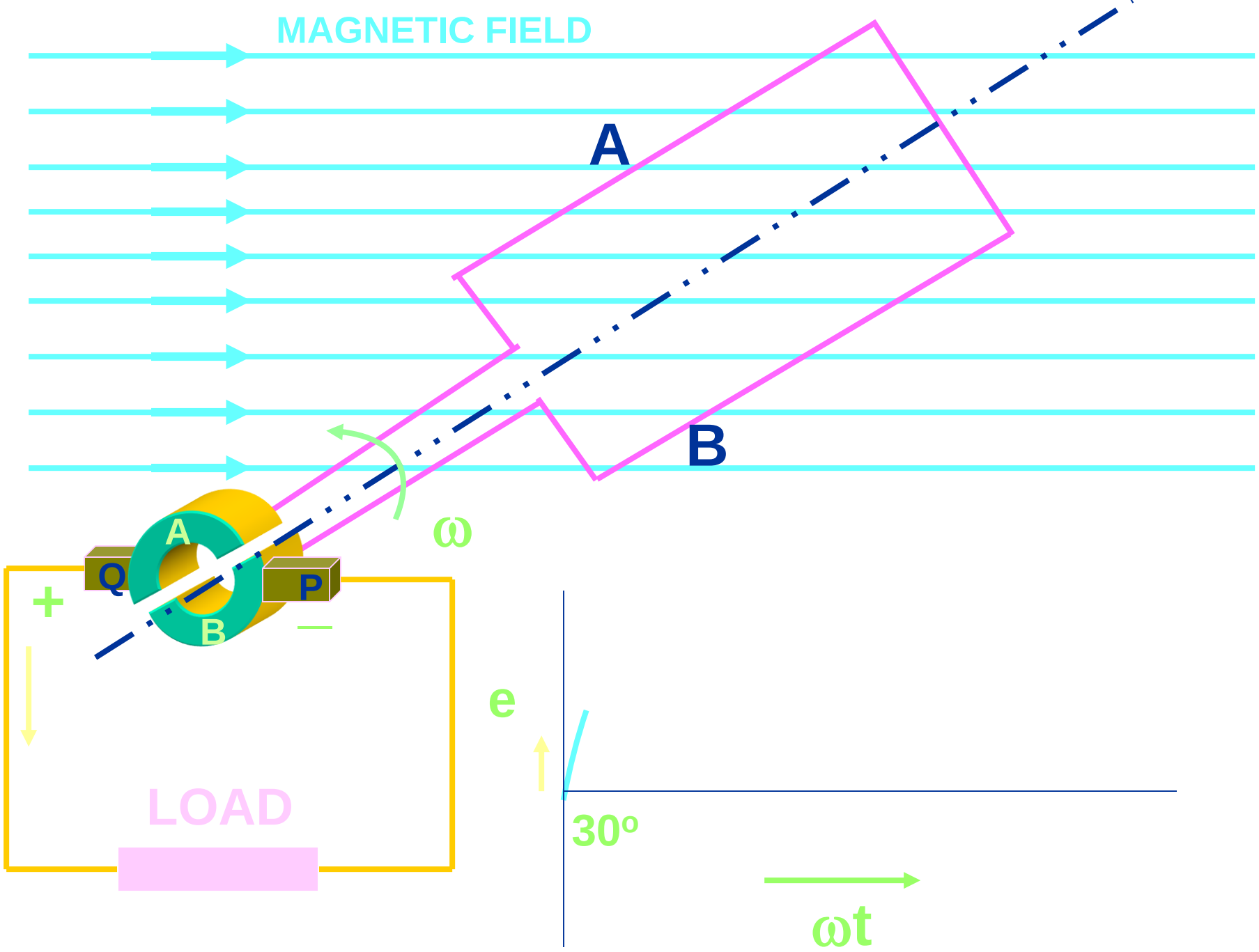
MAGNETIC FIELD
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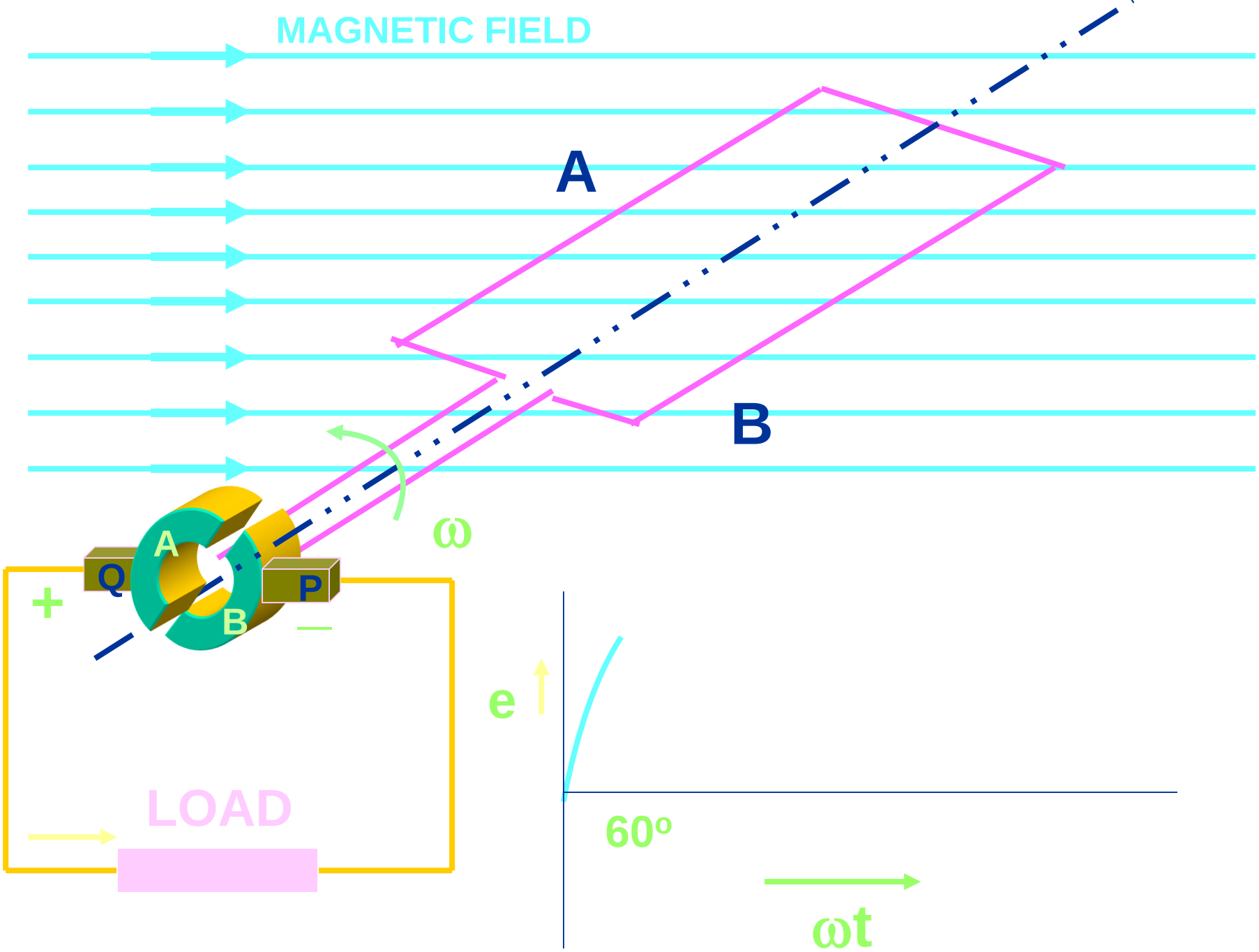
MAGNETIC FIELD
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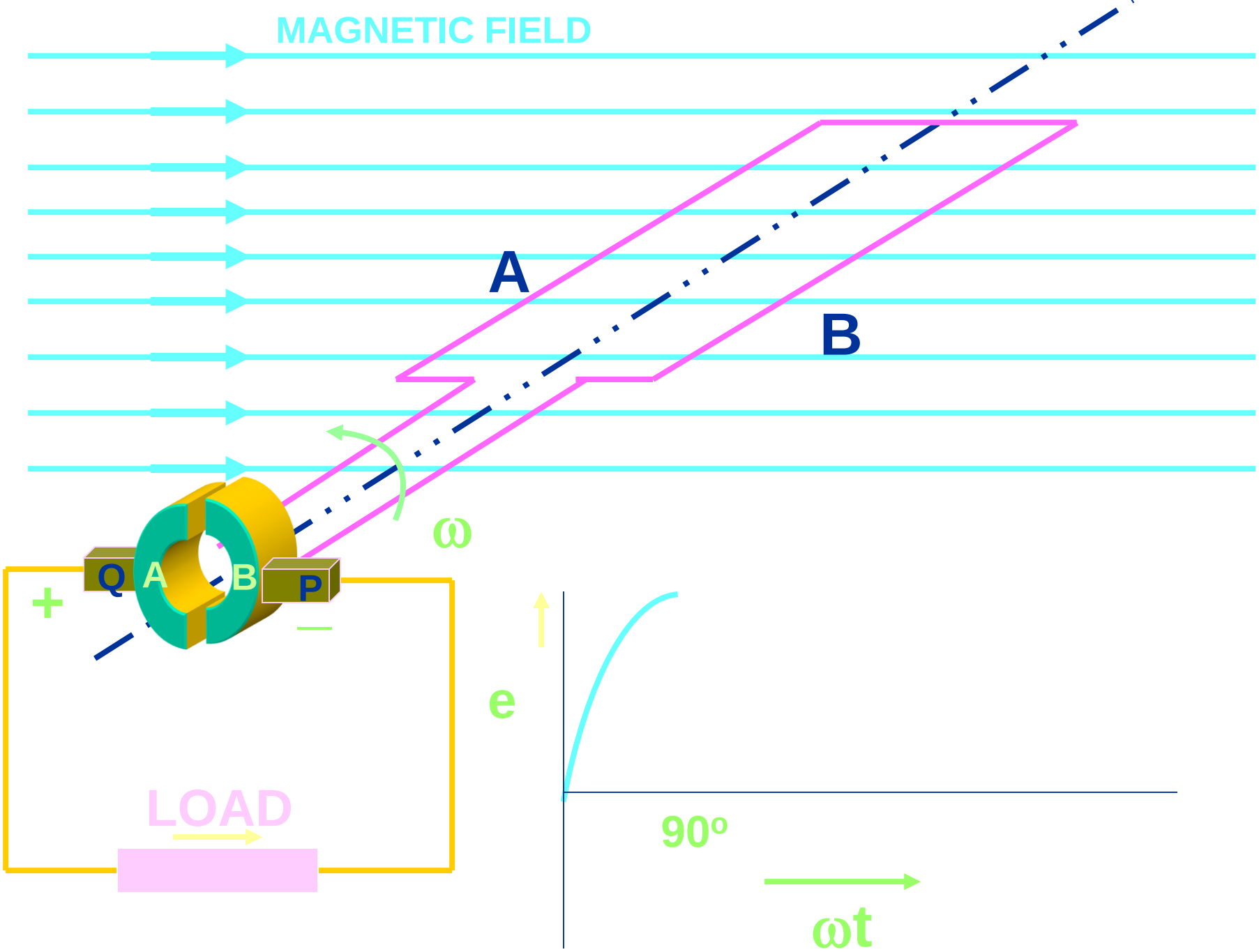
MAGNETIC FIELD
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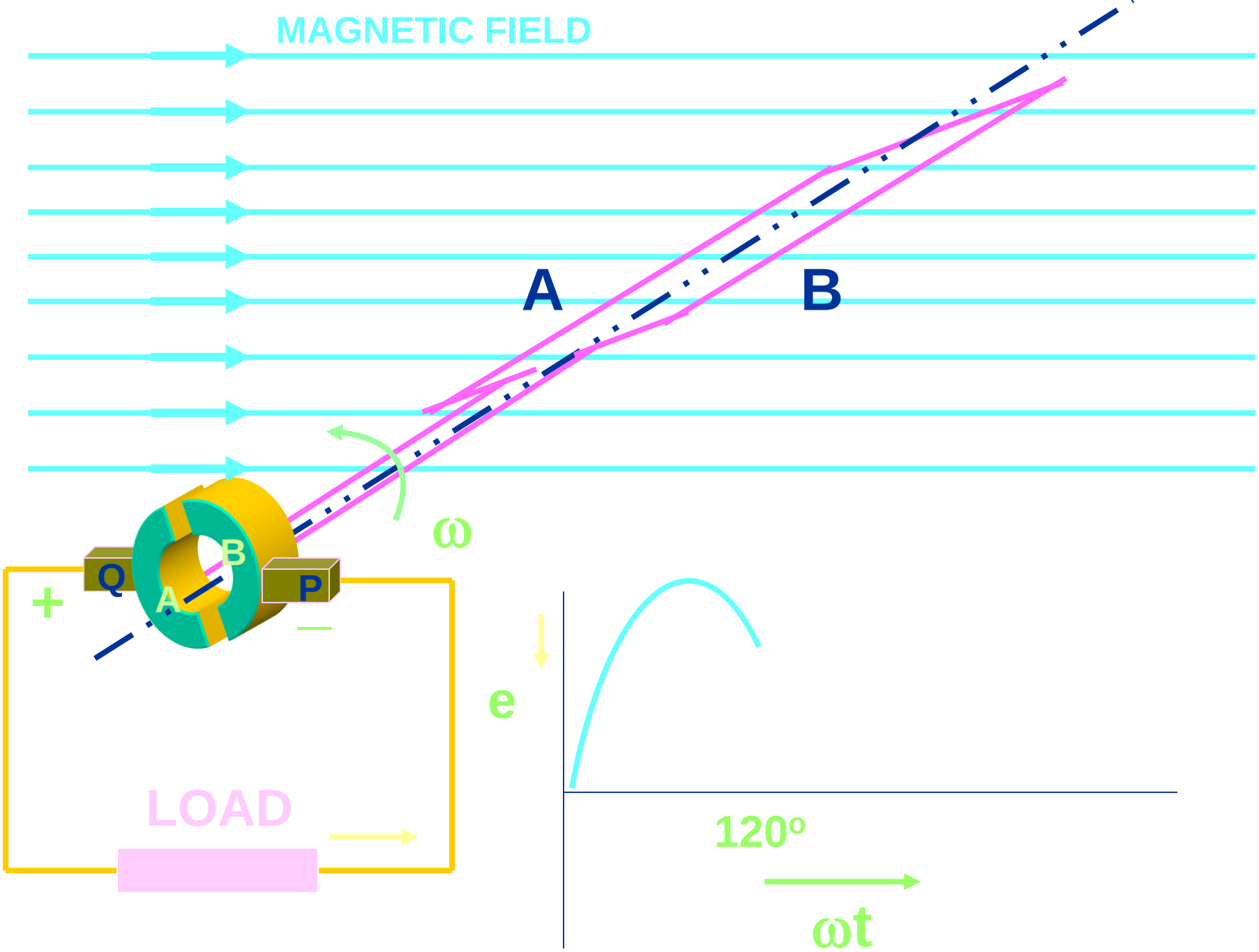
MAGNETIC FIELD
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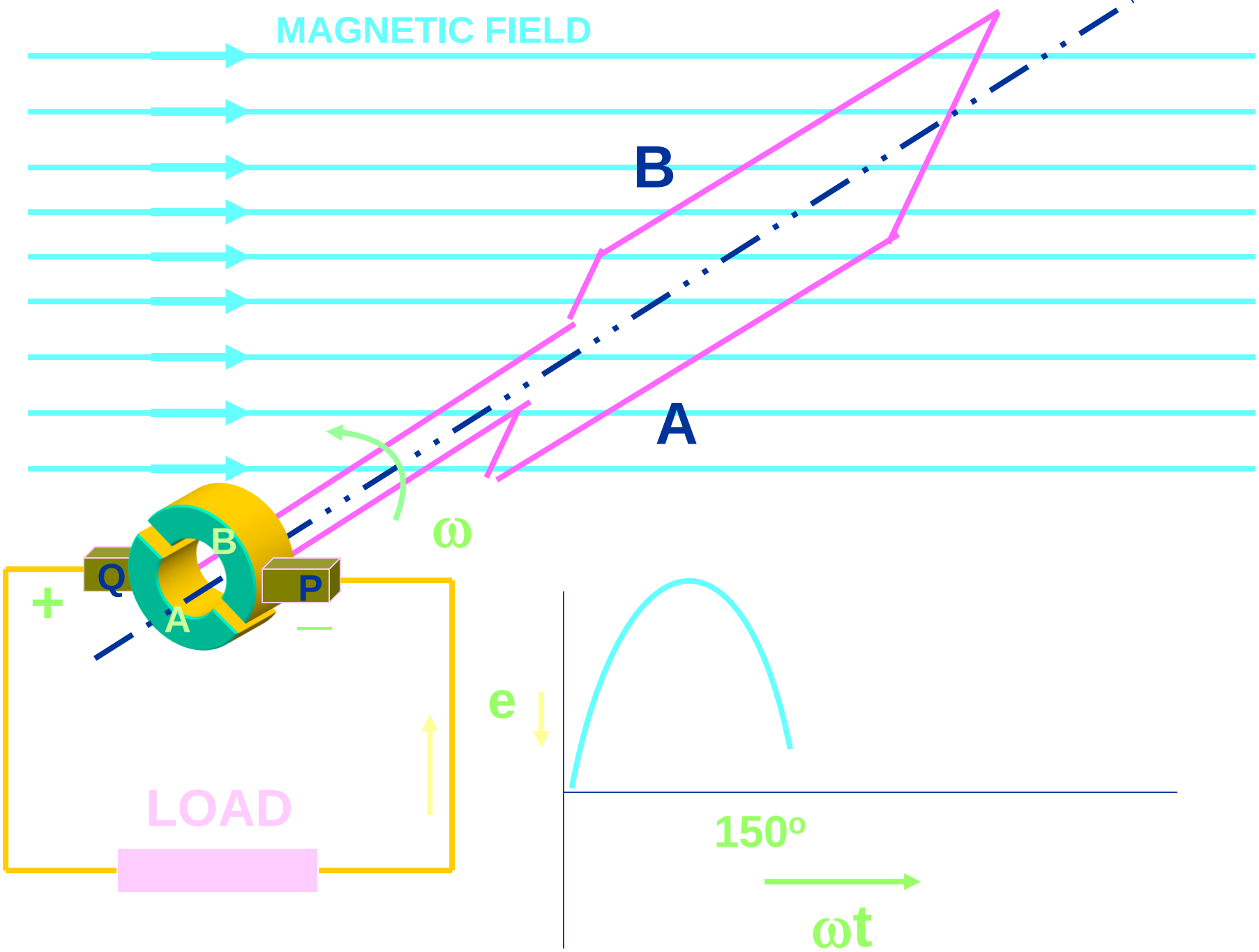
MAGNETIC FIELD
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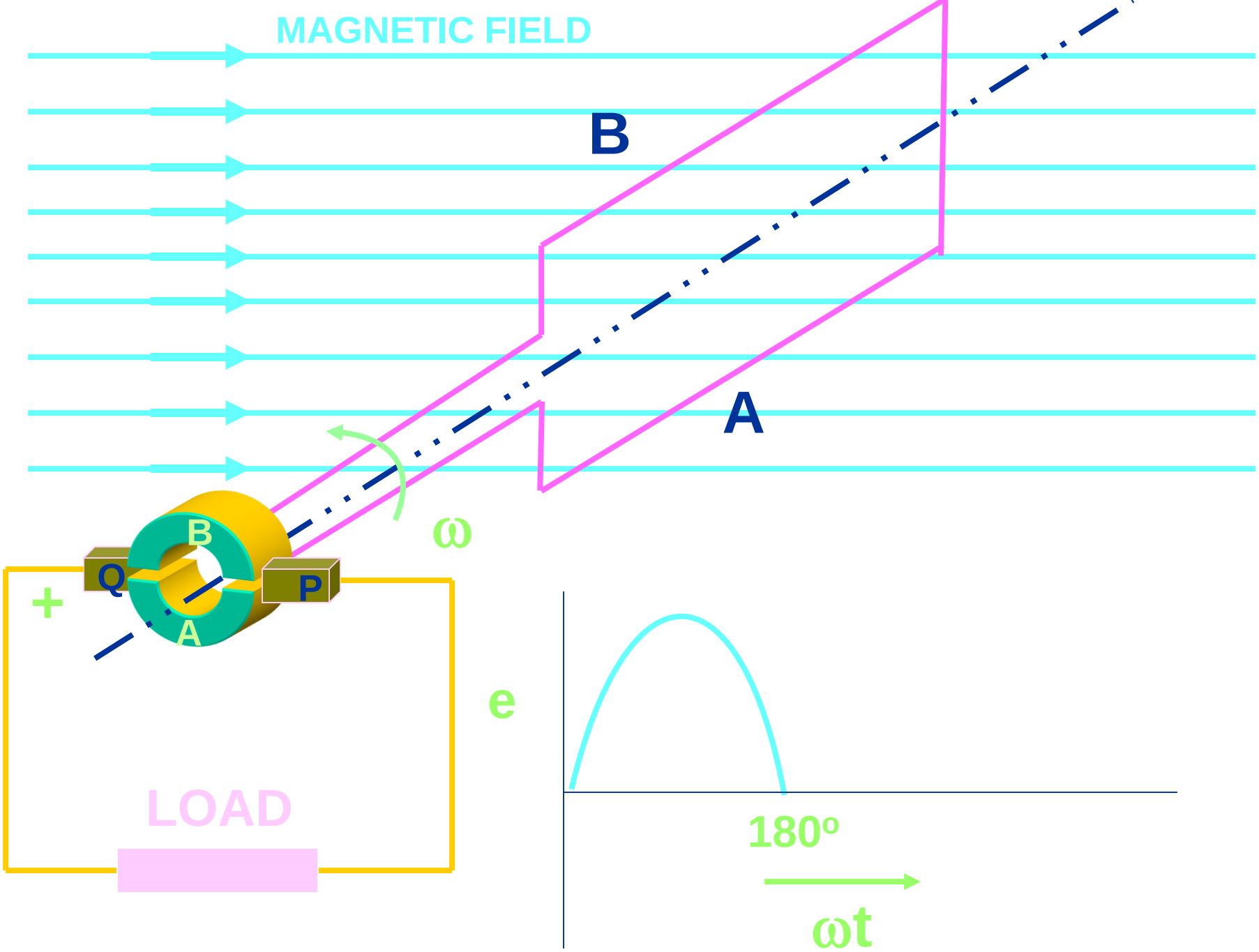
MAGNETIC FIELD
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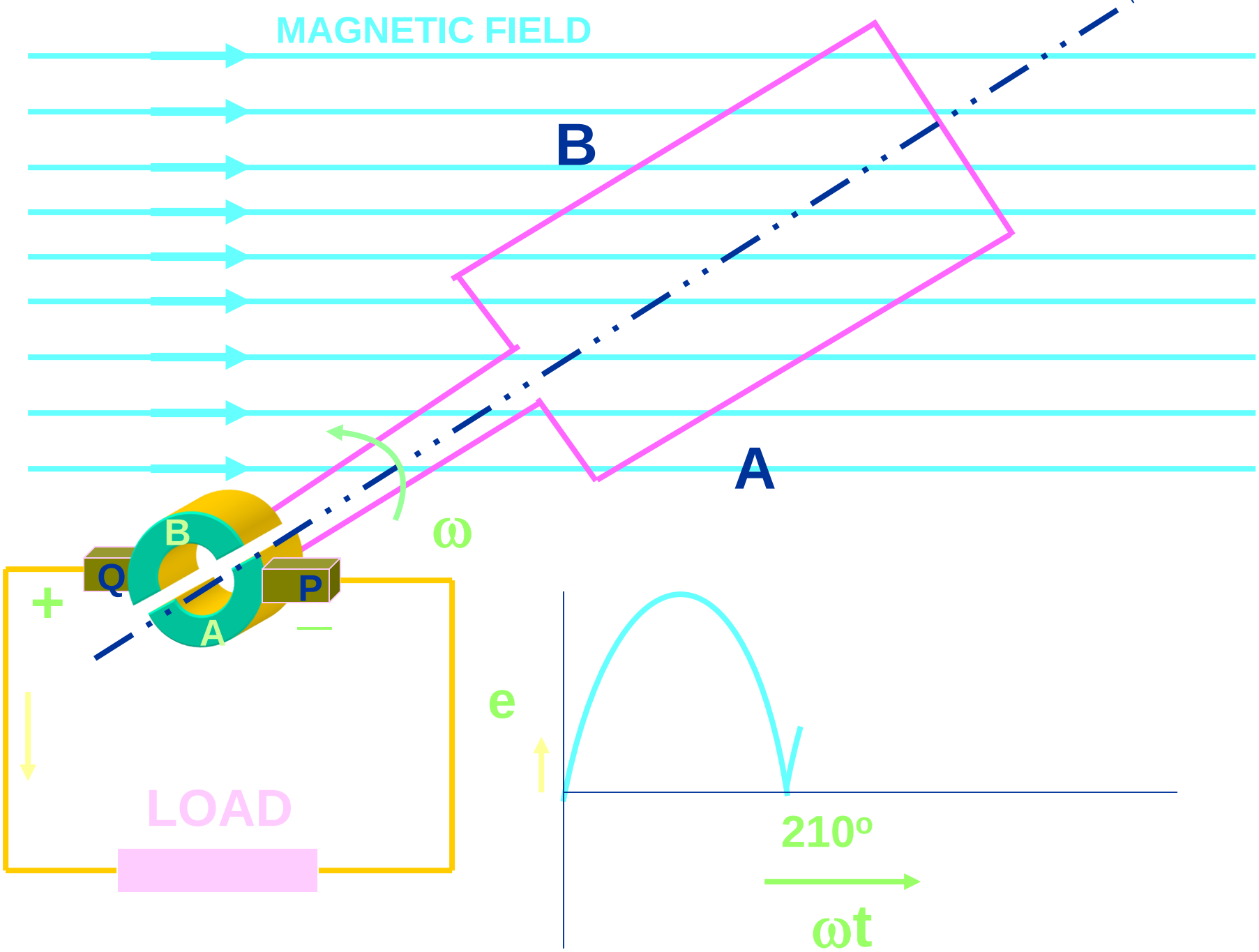
MAGNETIC FIELD
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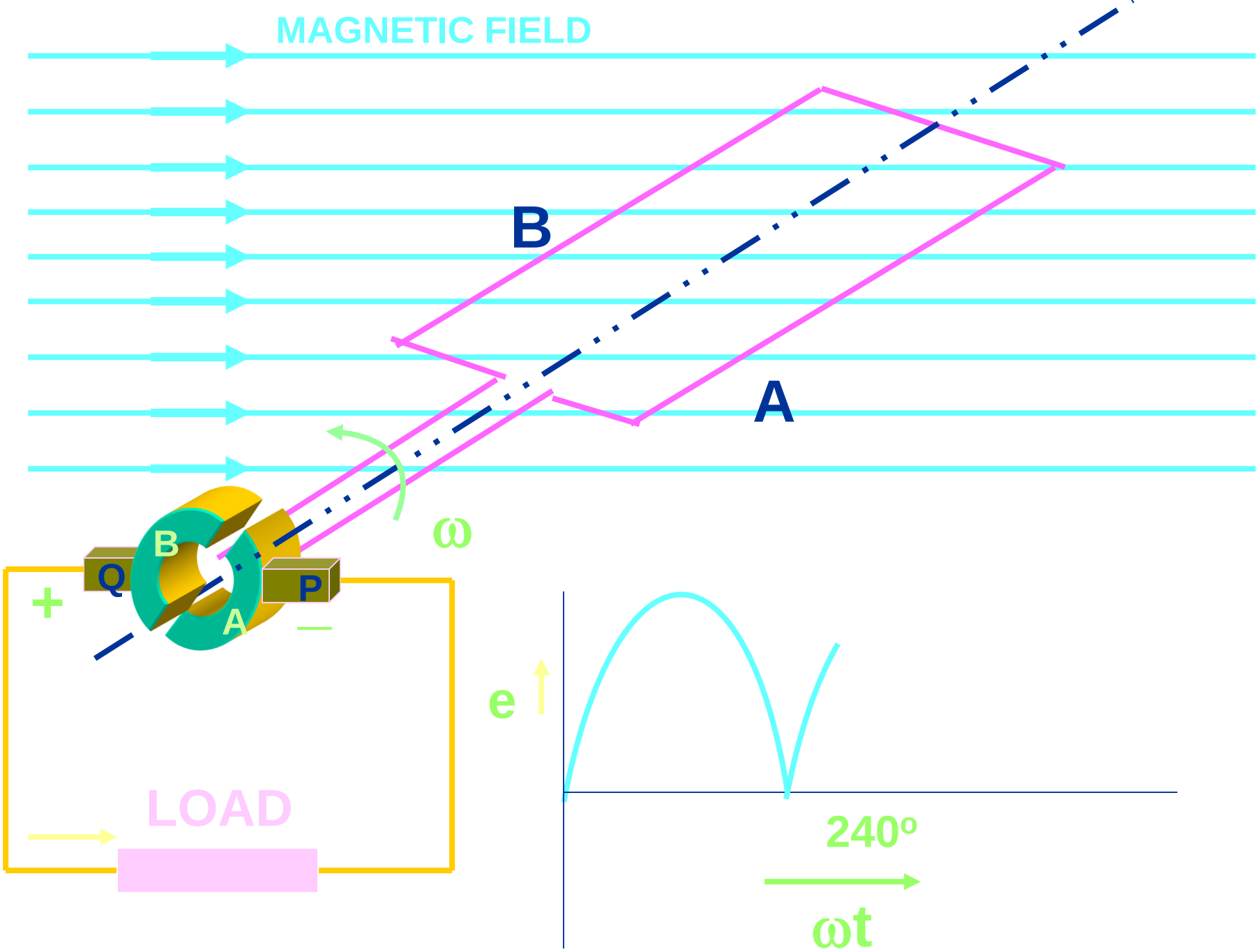
MAGNETIC FIELD
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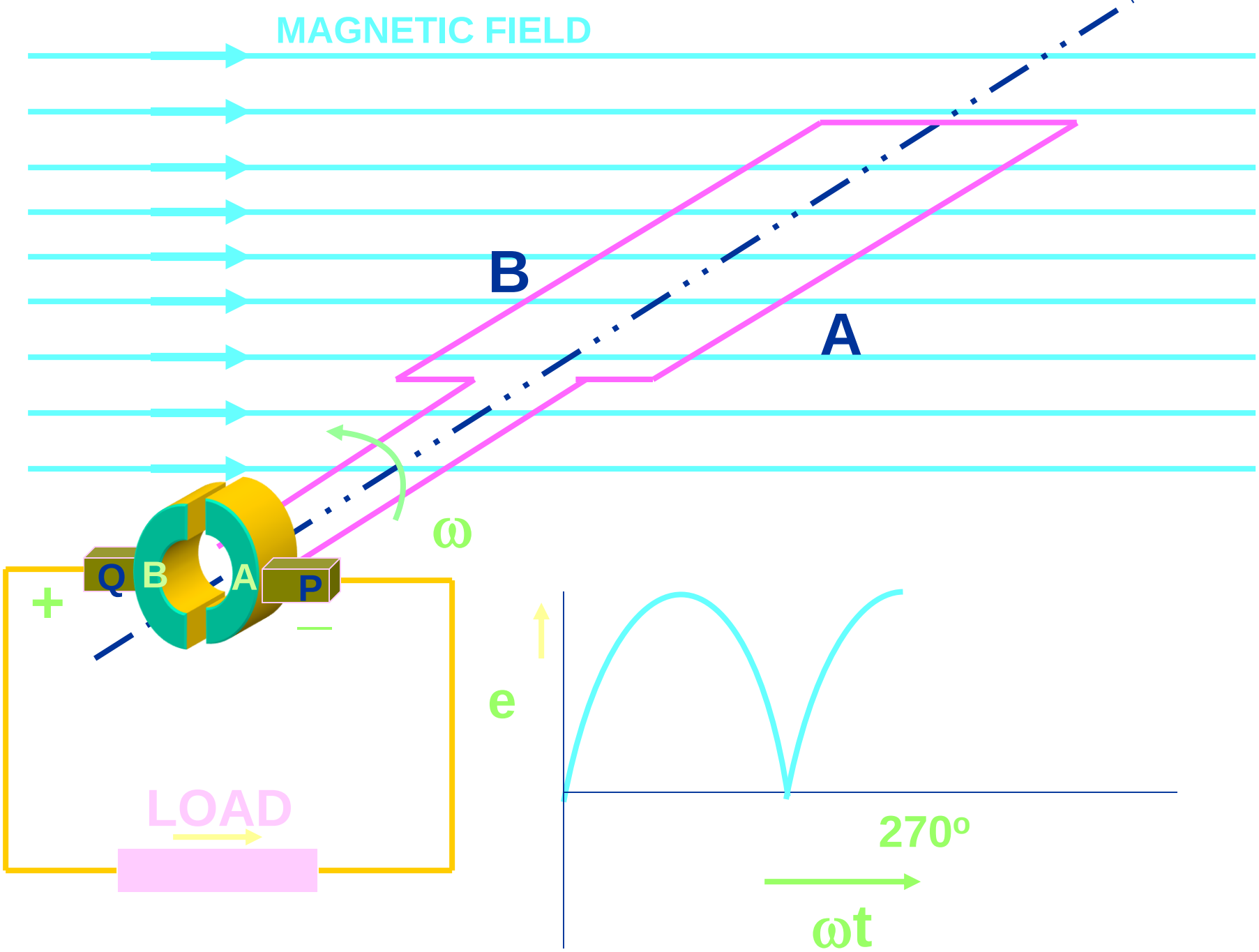
MAGNETIC FIELD
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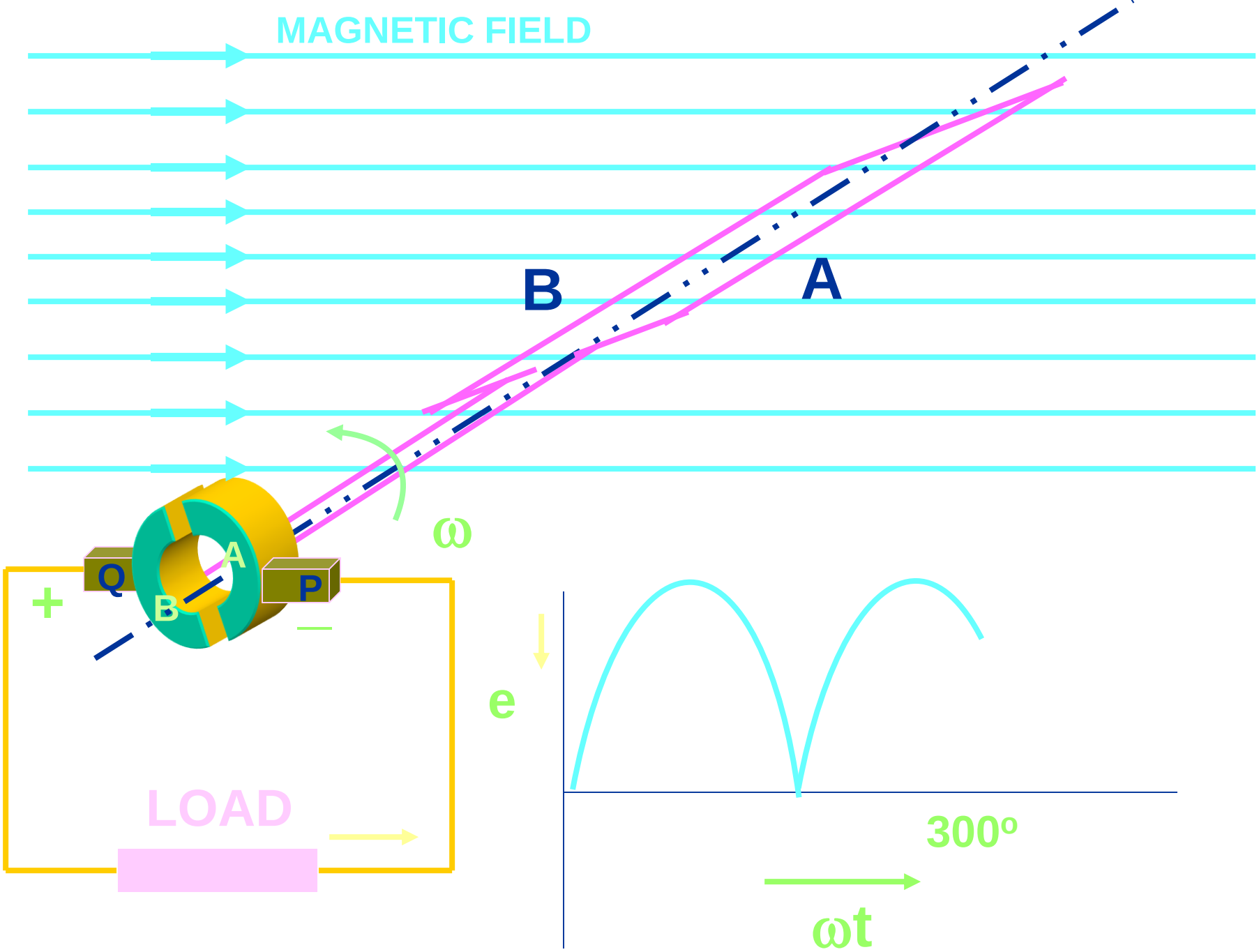
MAGNETIC FIELD
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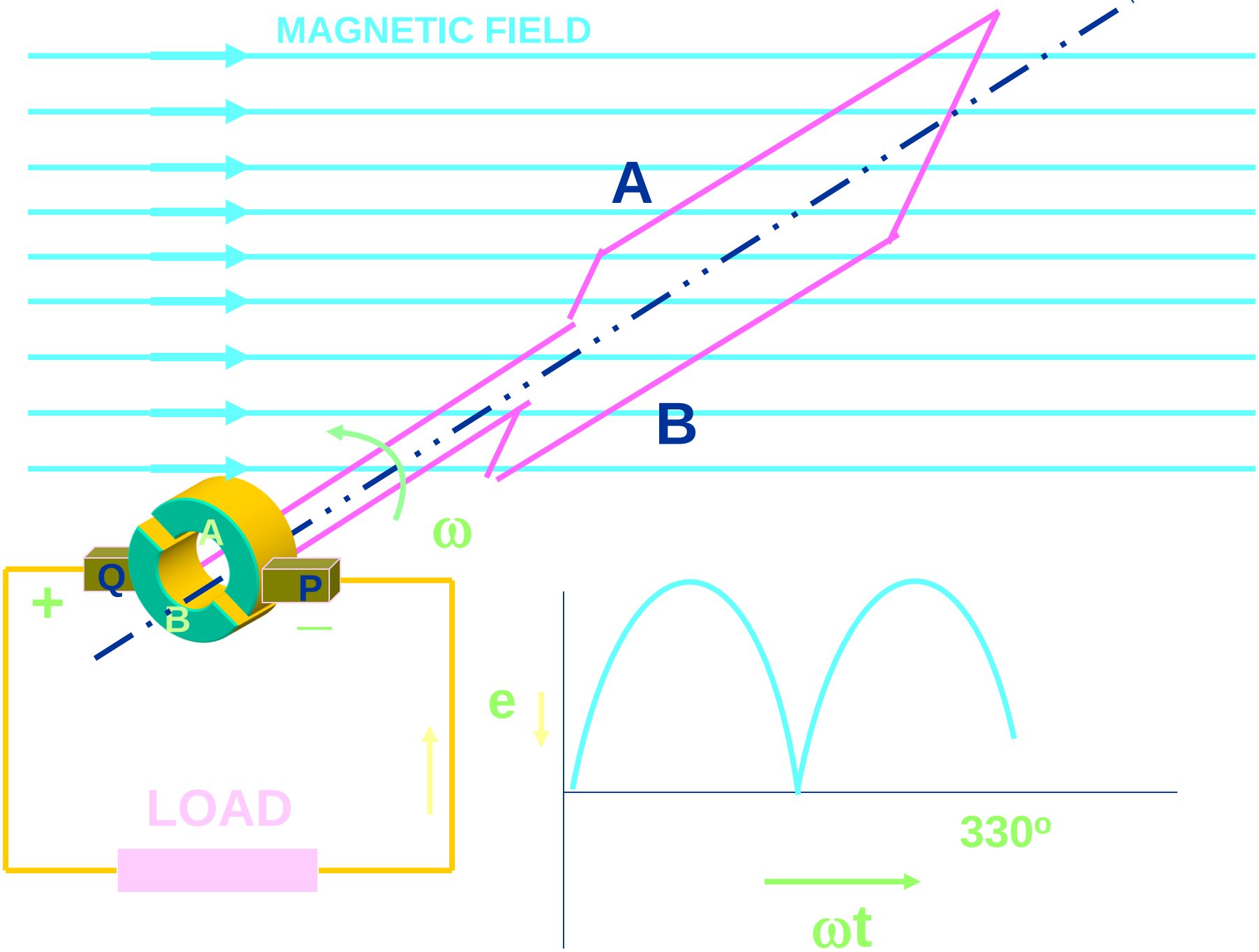
MAGNETIC FIELD
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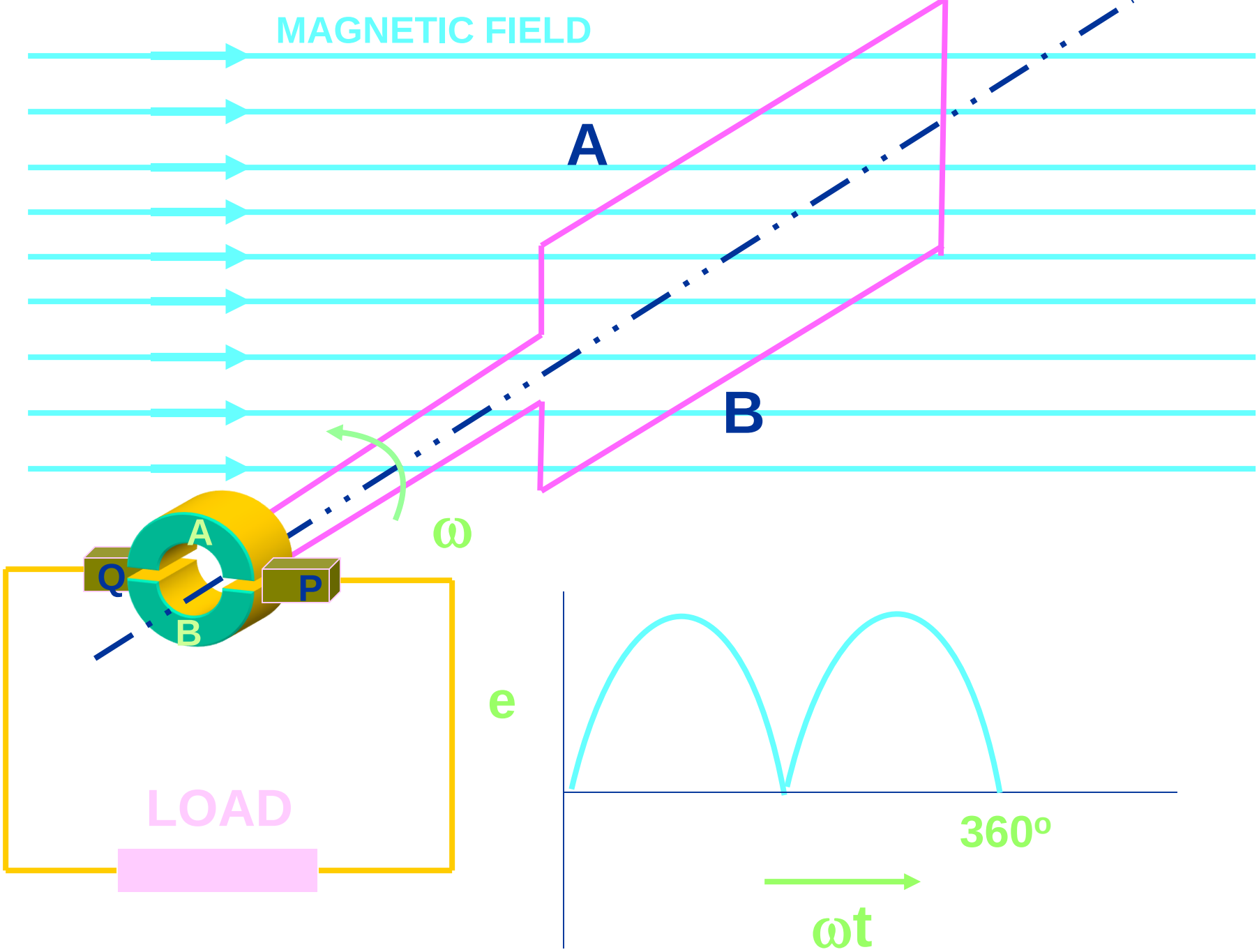
MAGNETIC FIELD
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Worked Examples:

A four pole generator having wave-wound armature winding has 51 slots, each slot containing 20 conductors. What will be the voltage generated in the machine when driven at 1500 rpm assuming the flux per pole to be 7.0 mWb ?

For a simplex wave wound generator,

$$Emf\ Generated\ /\ Path = \frac{\phi ZN}{60}\left(\frac{P}{A}\right)$$

$$E_g = \frac{P\phi ZN}{60A}$$

$$\phi = 7 \times 10^{-3} Wb$$
$$Z = 51 \times 20 = 1020$$
$$A = P = 4$$
$$N = 1500\ rpm$$

$$E_g = \frac{7 \times 10^{-3} \times 1020 \times 1500}{60}\left(\frac{4}{2}\right)$$

$$= 357V$$

Questions

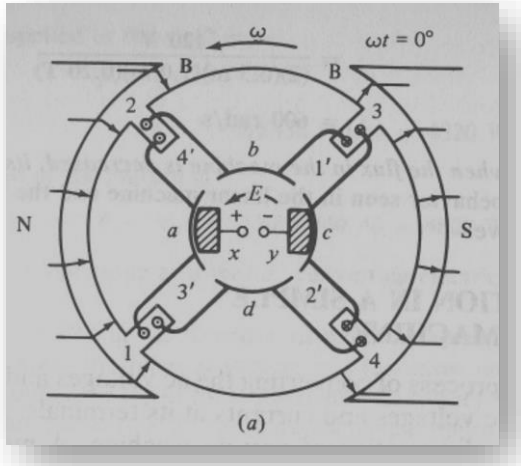
- Explain and describe using drawings the construction of dc machine
- What is the function of the following in dc machines:

  a- armature winding

  b- field winding
- Explain how dc machines can work as generator and motor
- State some applications and types of connections of dc machines (generator and motor)
- Derive the EMF equation for dc machines

Commutation

- **Commutation is the process of converting the ac voltages and currents in the rotor of a dc machine to dc voltages and currents at its terminals.**

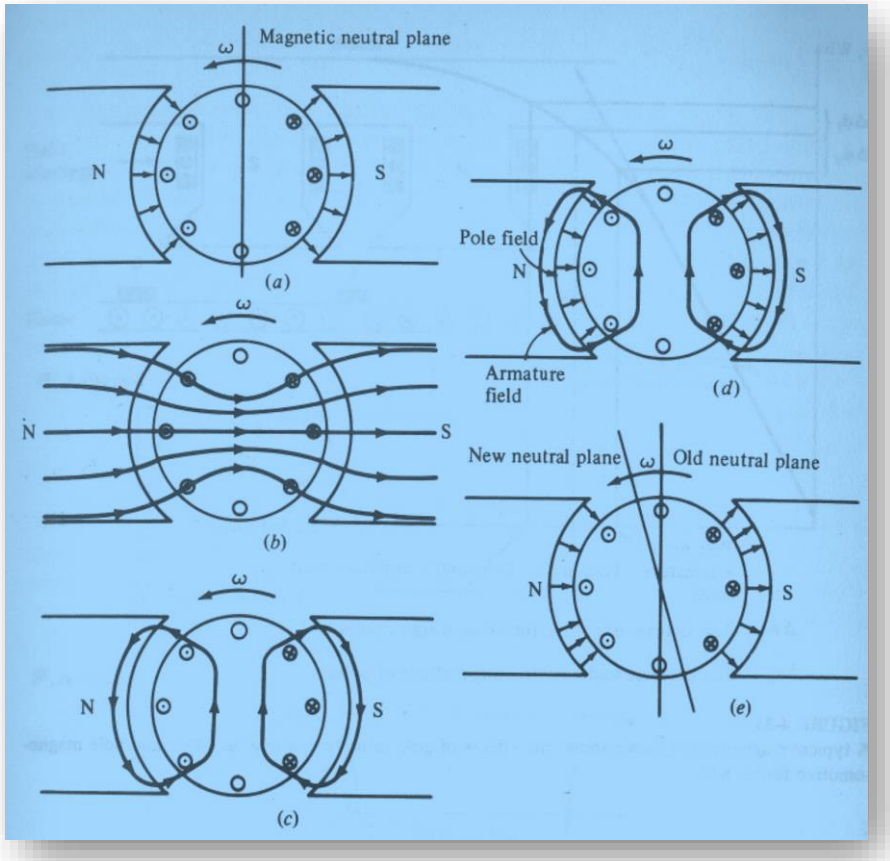


Problems with Commutation in Real Machines

- **Armature reaction**

The current though the armature conductors set up a magnetic field surrounding it which has the following effects

- **Weakens the main flux**
- **Distorts the main flux**
- **Neutral plan shift**



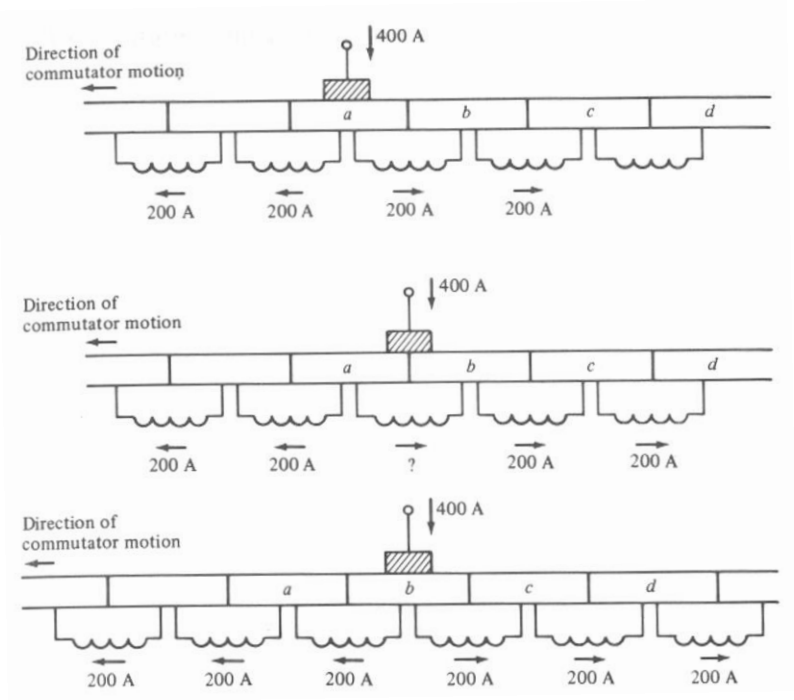
Problems with Commutation in Real Machines

- **L(di/dt) Voltage**

Occurs in the commutator segments being shorted out by the brushes > inductive kick

These effects causes

- Arcing and sparking at the brushes
- Flashover
  - Reduce brush life
  - Pitting of the commutator segment

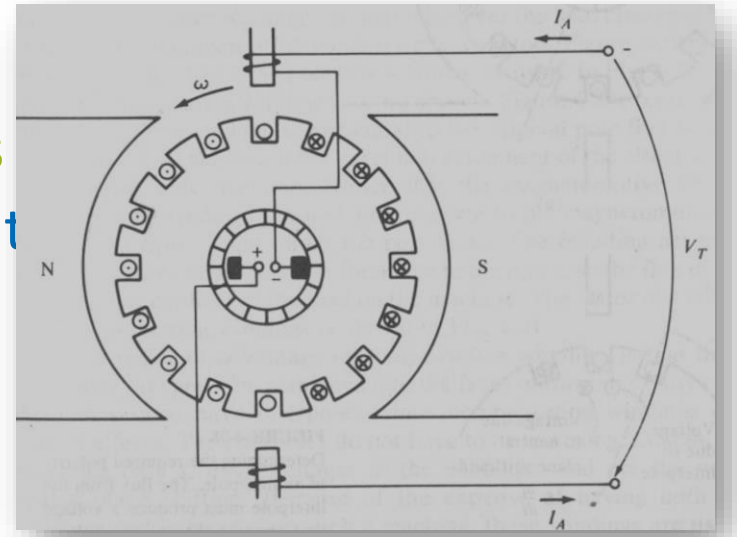


Solutions to Problems with Commutation in Real Machines

- **Brush shifting**
- **Commutating poles or interpoles**
- **Compensating windings**

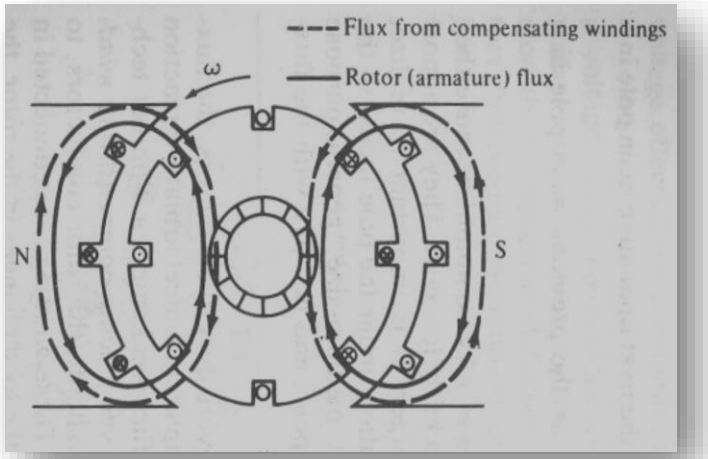
Solutions to Problems with Commutation in Real Machines

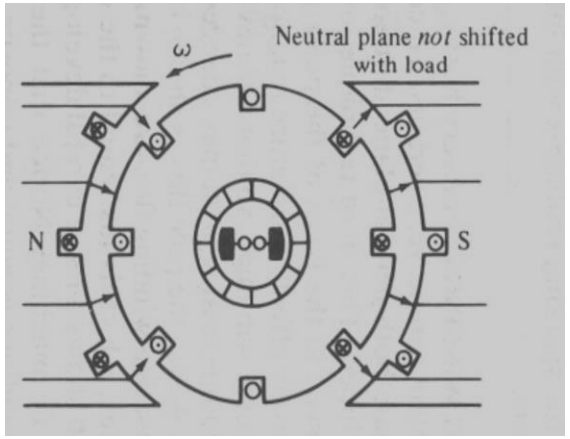
- **Commutating poles or interpoles**
- It cancels the voltage in the coils undergoing commutation
- interpole windings are in series with the rotor windings
- as the rotor current increases flux produced by interpole also inceases
- producing an oppssing effect to that of neutral plan shift



Solutions to Problems with Commutation in Real Machines

- **Compensating winding**
- Solves the problem of flux weakening and neutral plane shift
- Compensating windings are in series with the rotor windings
- placing in slots carved in the faces of the poles parallel to the rotor conductors

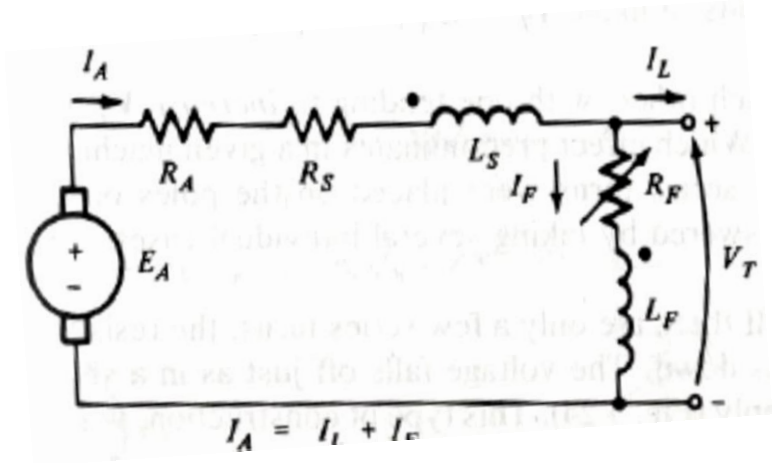


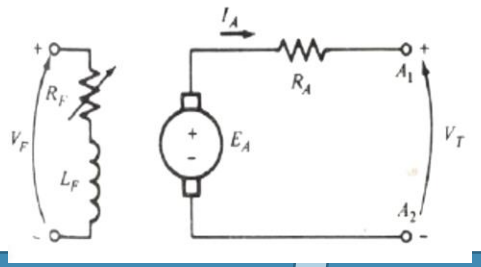


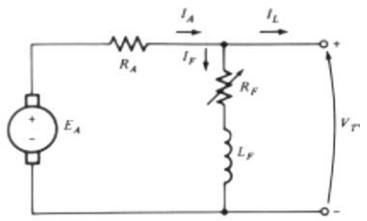
DC GENERATORS

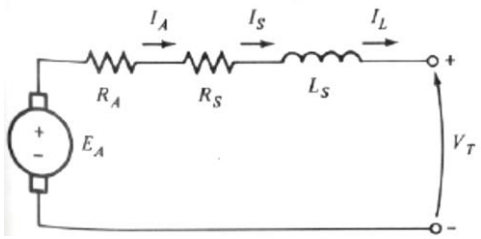
There are four major types of DC generators, namely

- Separately excited generator.
- Shunt generator.
- Series generator
- Compounded generator
  - Cumulative
  - Differential

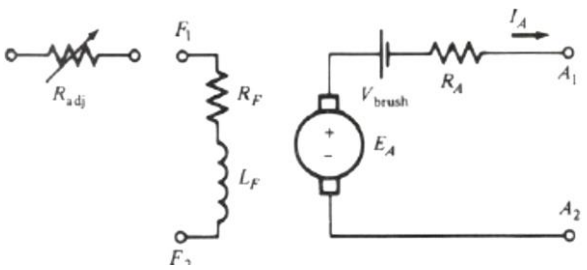








The Equivalent Circuit of a DC Generator



Two circuits are involved in DC generators

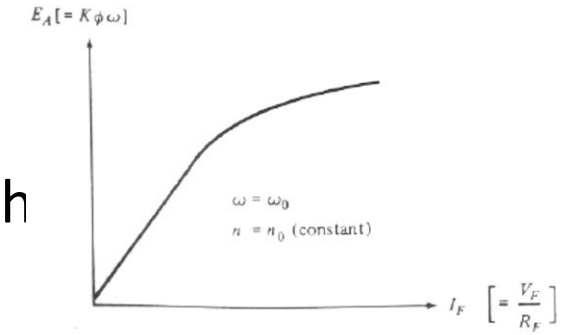
- **Armature Circuit**
- **Field circuit**

Armature circuit represents Thevenin equivalent of the entire rotor.

- It cantain an ideal voltage source E_A and a resistor R_A. .
- Brush voltage drop is represented by a small battery
- The field coils, which produce the magnetic flux
  - inductor L_F and resistor R_F
  - R_{adj} for field current control

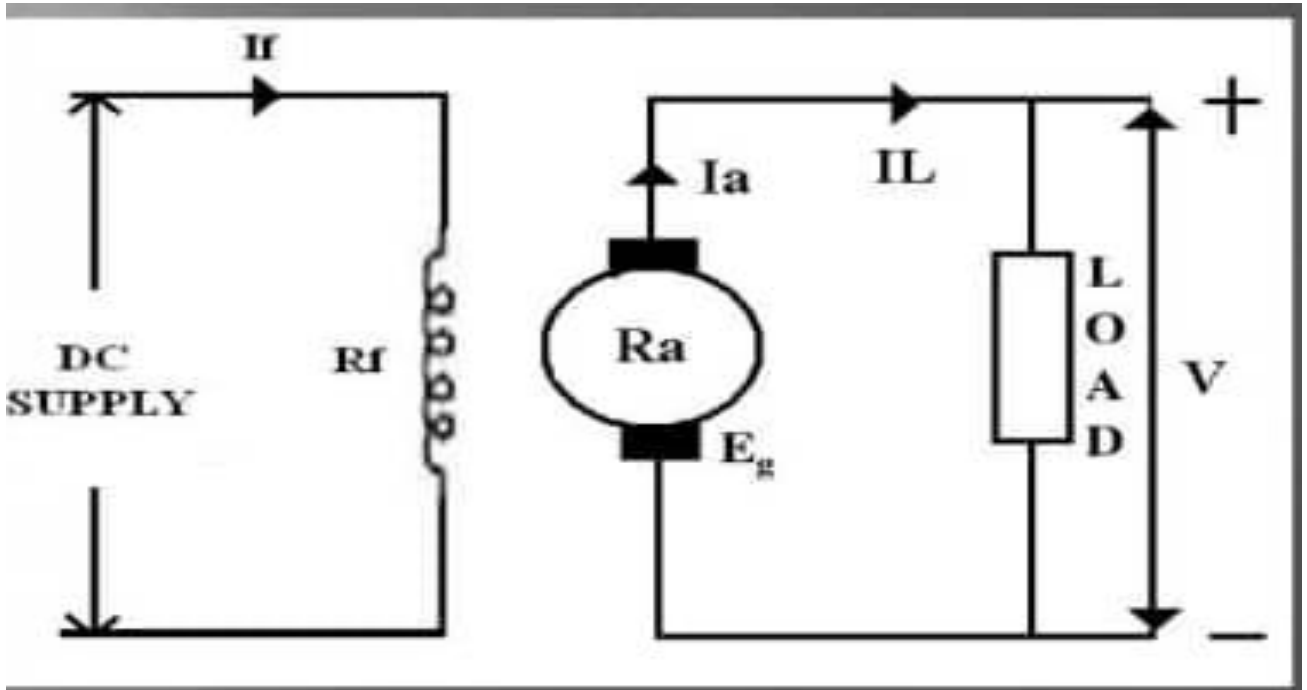
Magnetizing curve of a DC Generator & performance

- The internal generated voltage EA of a dc generator is given by $E_A = K\Phi\omega$
- E_A is directly proportional to the flux
- The field current is directly proportional to th magnetomotive force and hence E_A



- Brush voltage drop is represented by a small battery
- Performance of the DC generators are determined by terminal output parameter I_L and V_T
- Voltage regulation also determines its performance

Separately Excited DC Generator



Armature current, $I_a = I_L$

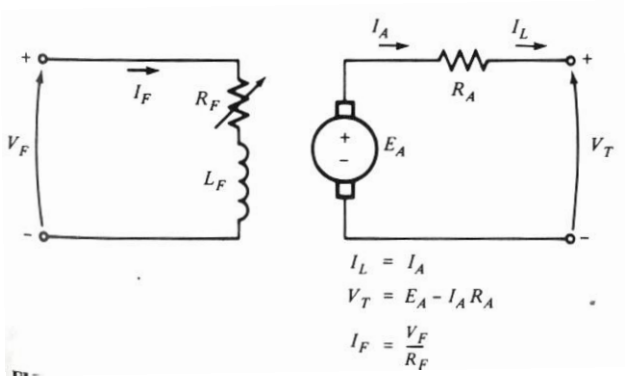
Terminal voltage, $V = E_g - I_a R_a$

Electric power developed $= E_g I_a$

Power delivered to load $= E_g I_a - I_a^2 R_a = I_a(E_g - I_a R_a) = VI_a$

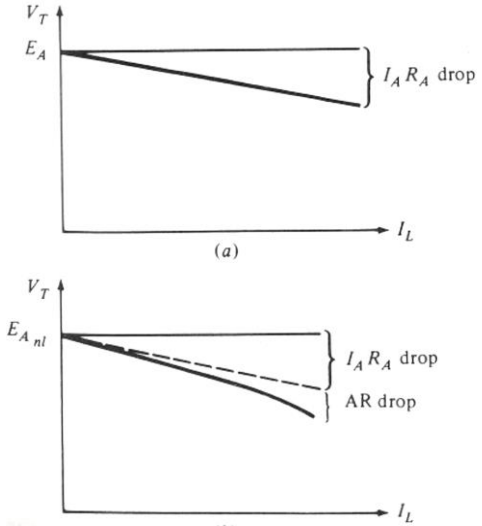
The Separately Excited Generator

- A separately excited dc generator is a generator whose field current is supplied by a separate external dc voltage source.



A separately excited dc generator

- By Kirchhoff's voltage law, the terminal voltage is $V_T = E_A - I_A R_A$
- Since the internal generated voltage is independent of lA the terminal characteristic of the separately excited generator is a straight line



The terminal characteristic (a) with and (b) without compensating windings

The Separately Excited Generator

Control of Terminal Voltage > two methods

- Change the speed of rotation
- $E_A = K\Phi\omega\uparrow$ >$VT = E_A \uparrow - I_A R_A$ > $V_T \uparrow$

- Change the field current.
- $I_F = VF/R_F\downarrow$ > $I_F \uparrow$ > $\Phi \uparrow$> $E_A = K\Phi\uparrow\omega$ >
- $V_T = E_A \uparrow - I_A R_A$ > $V_T \uparrow$

The terminal characteristic (a) with and (b) without compensating windings

Problem 1.

A dc machine (6 kW, 120 V, 1200 rpm) has the following magnetization characteristics at 1200 rpm.

I_f	(A)	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.8	1.0	1.2
E_a	(V)	5	20	40	60	79	93	102	114	120	125

The machine parameters are Ra =0.2 Ω, Rf =100 Ω. The machine is driven at 1200 rpm and is separately excited. The field current is adjusted at If =0.8 A. A load resistance RL =2 Ω is connected to the armature terminals. Neglect armature reaction effect.

(a) Determine the quantity KaΦ for the machine.

(b) Determine Ea and Ia.

(c) Determine torque T and load power PL.

Repeat Problem 1 if the speed is 800 rpm.

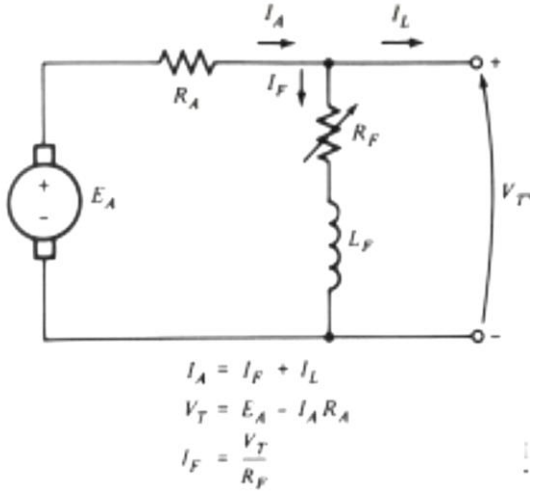
The Shunt Generator

A shunt dc generator is a dc generator that supplies its own field current by having its field connected directly across the terminals of the machine.

- The armature current of the machine supplies both the field circuit and the load

$$I_A = I_F + I_L$$

- $$V_T = E_A - I_A R_A$$
-

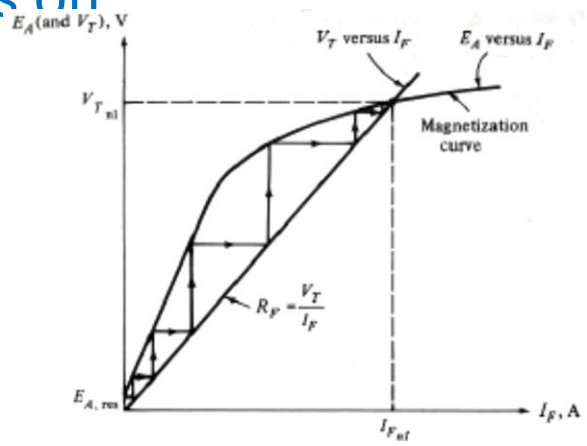


The equivalent circuit of a shunt de generator

The Shunt Generator

Voltage Build up in a Shunt Generator depends on

- Residual flux
- $E_A = K\Phi_{res}\omega$
- $I_F = V_T \uparrow / R_F > E_A = K\Phi\uparrow\omega >$
- $V_T = E_A \uparrow - I_A R_A > V_T \uparrow$



Voltage buildup on starting in a shunt dc generator

possible causes for the voltage to fail to build up during starting

- There may be no residual magnetic flux
- The direction of rotation of the generator may have been reversed
- The field resistance may be adjusted to a value greater than the critical resistance

EXCELLENT

THANK
YOU